

Fig. 5: Methods classification

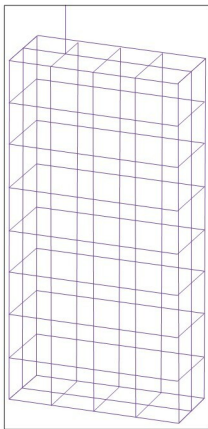


Fig. 6: Wire-frame MOM model of a cellular telephone handset with an antenna connected to the mesh (Courtesy: Book by Thomas A. Milligan)

the resonant frequencies and propagation modes of the waveguide, cavity, propagation constant, etc. The results are field configurations that would be preferred to be excited by real environmental sources.

Ray tracing (or optics). As the electrical size increases, the integral and differential methods become useless.

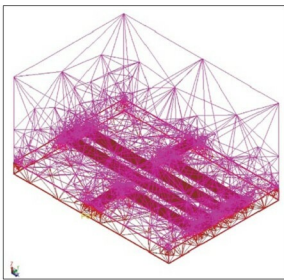


Fig. 7: Typical tetrahedral mesh used in FEM simulation (Courtesy: Agilent Technologies)

Then the optics method combines the Green's function to calculate fields by current and then uses the boundary conditions to determine current induced on the objects. This domain has two methods: PO and UTD.

Methods using integral Maxwell equations

Method of moments (MOM). Also known as 3D planar method, it uses Green's function over the planar structure, thus reducing the Maxwell equations to algebraic expressions. Layered stack up-structures can also be analysed as the Green's function can be applied, thus x-y plot is established as in planar structure. It actually approximates the current distribution as

sum of basic functions. In this method the calculations are required only at boundary values and not throughout the space, as shown in Fig. 6.

MOM creates a surface mesh over the modelled surface. MOM solutions are typically limited to objects that are only one or two wavelengths in size, for example, to calculate and extract detailed multiport s-parameter of all the interconnects on PCBs. Analysis of large structures becomes difficult because of the large amount of memory and time required to compute.

Some disadvantages of the MOM method are:

1. As the geometry increases, the storage requirement and computing time increase as the square of the size of geometry.
2. It can be applied only to those problems for which the Green's function can be calculated.
3. Different frequencies require different formulations (due to frequency domain).
4. It does not satisfy boundary condition at every point; satisfies only over an integral average of boundaries.

Multilevel fast multipole method (MLFMM). It is an alternative approach to the MOM as it requires less memory. It is suitable for analysis of very large problems, such as RCS and reflector antenna. According to emfieldsolution.com, "The object is placed in a box which is then split into eight smaller boxes. Each of the boxes is then divided again recursively until the size of smaller boxes contains few basic functions." It also illustrates that, if a particular MOM solution requires 150GB memory then MLFMM would require only 1GB.

Partial element equivalent circuit (PEEC). In this method the integral equations are interpreted as Kirchhoff's voltage law applied to basic cell. So, overall, the geometry in this